

Lesson 4 — Pseudocode — Plan Before You Build

Matches LEGO Teacher Portal lesson: "Dance Designer".

Objective (I can...): I can write my plan in plain words (pseudocode) and turn it into blocks. **CS concept:** Pseudocode / algorithm planning. Writing the plan in plain words is a first taste of **abstraction — hiding the messy details** so you can think about the big steps first. · **Builds on:** Lesson 3 (decomposition).

AI Connection (2 min): Engineers write AI's behavior in plain words first ("IF you see a face, THEN wave") before real code. *First-principles question:* "Why plan in words before building?"

Materials: LEGO CS & AI kit + device per pair, a "plan sheet" per pair, a stack of blank "mission cards."

Model reuse: same Basics model from Lesson 2 — code plus one tiny build tweak.

60-minute flow - Warm-up (8): "Secret mission cards" — each pair writes a short plain-word plan (3–4 lines) for a behavior on a mission card, then trades cards with another pair. Your card is the *other* pair's mission to build and run. Clear plain words win the mission. - **Concept + demo (10):** Coach writes 3–4 plain-word lines, then matches each to a block live — pseudocode is the bridge from idea to code. (*Rotate the format: try "coach makes a mistake, kids catch it" with one mismatched line.*) - **Build/Code (30):** Pairs build the mission card they received: write/refine pseudocode on the plan sheet, then code it line-by-line on the built model and run to check it matches. **Make-it-yours beat:** add or swap one brick so the bot fits the mission's theme. *Swap at 15 min.* - **Challenge (7):** Change one plan line (e.g., a longer wait) and update the matching block, keeping plan and code in sync. - **Share + pack-up (5):** Pairs read the mission pseudocode aloud, then run to prove the match. Model stays built.

Pair roles: Navigator owns the plan sheet; Driver translates lines to blocks. *Swap at 15 min. (Trio: third child checks each line was coded.)*

Differentiation: *Support* — sentence-framed pseudocode ("First the model . **Then it** ."). *Extension / fast-finisher* — write 6+ pseudocode lines. *Reteach* — code one plan line together, then let them do the next alone.

Exit check: Pair's pseudocode lines line up one-to-one with their working blocks. **At-home (optional):** Write plain-word pseudocode for getting ready for school (5 lines). ~10 min.